Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Particular integral of standard functions - Type 1

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Particular integral of standard functions - Type 1

1 Particular integral of standard functions - Type 1

Study focus: Particular integral of standard functions - Type 1. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000046 · EV0000047

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Methods to solve a non-homogeneous linear differential equation of the type f of b of y is equal to capital X when capital X is equal to E par ax.

Key details

- In the previous session, we learned methods to solve.
- Homogeneous linear differential equation with constant coefficients.
- We know that the general solution of the non-homogeneous LDE with constant coefficient has two components, complementary function and particular integral.
- And complementary function is the general solution of the associated homogeneous differential equation.
- We already learned methods to find complementary function.
- Basically, in this session, we are finding methods to find.
- The particular integral of the differential equation f of df y is equal to e power ax.
- What is this inverse differential operator?
- Why do we need this inverse differential operator?
- The inverse differential operator is denoted as 1 by eight.
- f of d.
- 1 over f of d and 1 and f of d are inverse operators and f of d of 1 over f of b of x is equal to x itself.
- They are inverse operators.
- You can change the order.
- You can also write it as 1 over f of d of f of b of x is equal to capital X.
- over E f of D or 1 over Af of B of X.
- It's very important statement.

- F of B of Y is equal to capital X.
- It is a non-homogeneous linear differential equation.
- Just recall what is a non-homogeneous LD?
- If RHS of the differential equation is a non-zero function of the independent variable, then the differential equation.
- Is called as a non-homogeneous linear differential equation.
- Is equal to x itself.
- Why it is equal to x?
- Because by definition of 1 over f of d, 1 over f of d of x is that function of x, which when operated upon by f of d gives back x.
- So y is equal to 1 over F of D of capital X satisfies the equation f of d of y is equal to capital X, which means that y is equal to 1 over F of D of X.
- Is a solution of the differential equation.
- And this solution is called particular solution of the differential equation f of d of y is equal to capital X because this 1 over F of D of X do not contain any R.

Example or application from the lecture

- First, let us learn a very important concept called inverse differential operator.
- Let us prove this.
- Let us prove this statement.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 05:00–09:59

If you have a non-homogeneous differential equation here of Df y is equal to capital X, we know that the general solution of this non-homogeneous equation has two components.

Key details

- One is called complementary function, which is abbreviated as CF, and the other.
- It is the complete solution of the corresponding homogeneous differential equation.
- Now the next question is how do we find 1 over F of D of capital X?
- What is the meaning of this?
- It is 1 over F of D of capital X when F of D is equal to simple capital D.
- with respect to x without writing the constant of integration.
- 2 times integration of x without writing the constant of integration.
- 1 over D minus Ef capital X, so it is 1 over F of D of capital X, where F of D is the linear factor d minus E.
- What is 1 over d minus a f capital X?
- Replace a by minus a in the formula.
- Integral, E power X, X, Dx.
- It can be simplified in two ways.
- One way is applying the operator one by one.
- First you apply the operator 1 over d minus b on x.

- And you then you apply the operator 1 over d minus e on the resultant function of x.
- You get a function fx.
- Then on the resultant function you apply the operator 1 over d minus a.
- This is one way of simplifying 1 over d minus a into d minus b of x.
- The other way is resolve this 1 over d minus e into d minus p into partial fractions.
- Then use this formula and simplify.
- This 1 over f of d is an operator which is acting on the function x.
- It is not x into 1 over f of b because this is an operator which is acting on the function x.
- What if f of d is a cubic equation, cubic expression, then f of d would be product of three linear factors.
- Then this method is very then the now we are discussing methods to find particular integral when capital X, which is the right hand.

Example or application from the lecture

- Component is particular integral which is given the notation PI.
- We learn methods to find complementary function and the particular integral is given by 1 over f of d of capital X.
- So let us first learn some results, important results wherein f of d takes a simple function of D.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 10:00–14:59

Side function of the differential equation is a specific function of x.

Key details

- The first case being.
- When capital X is equal to E par ax.
- Basically, we are now learning methods to find the particular integral of e par ax, an exponential function a par ax.
- is e par ax by f of a provided f of a not equal to 0.
- It is 1 over f of b of e po is e power a x by f of a.
- In f of d, just replace d by a to get f of a.
- Provided f of a not equal to 0, the denominator not equal to 0.
- It is e power minus ax by f of minus a in the linear differential operator re.
- Place d by minus a if you have e par minus a x as the function.
- The cases of failure.
- When is f of a equal to 0?
- When d minus a is a factor of f of d.
- Then f of a is equal to 0, then a becomes a root of the equation f of d and f of a is equal to 0.
- When f of a is equal to 0, the formula is 1 over f of d of e power e x is equal to x into.
- E par a x by f dash a provided f dash a not equal to 0.

- See x has to be written outside the operator.
- It is x into 1 over f dash d of e par ax.
- And in f dash d replace d by a.
- You have f of d differentiate with respect to d to get f dash d.
- Then in f dash d put d is equal to a to get f dash a.
- C please understand x has to be written outside the operator x to the power 1 into e power a x by f dash a.
- What if f dash a equal to 0?
- When d minus a and d minus a two linear factors of f of b.
- into e par a is by f double dash a provided the denominator f dash double dash e not equal to zero.
- You differentiate this f of d two times with respect to d and then substitute d is equal to a in the second derivative to get f double dash a so provided this is not equal to 0.
- What if this f double dash a is equal to 0?
- The formula is x cube into e power e x by f triple dash a and so.
- But the same formula is applicable for the following functions.
- So, what is the first one? x is equal to k constant.
- D by 0 it is k by f of 0.
- 0 is not a root of the equation f of d is equal to 0.
- If 0 is a root, then it is x in 2.
- K by f dash 0.
- If f dash 0 is also 0, x square into k by f double dash 0 and so on.
- Now, so the same formula is applicable to next sine hyperbolic eggs.
- What is sine hyperbolic A in terms of exponential function?
- It is sine hyperbolic A is E par Ax minus e power minus e x by 2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 15:00–19:59

minus e power minus a x by f of d.

Key details

- Minus e par minus e x by f of b.
- In the first term we place d by a and in the second term replaced d by minus a.
- A now another function cos hyperbolic x.
- What is cos hyperbolic x? e par e x plus e par minus e x by 2.
- And what is cos hyperbolic x by f of d?
- It is half into e par a x by f of a plus.
- A power minus a is by f of minus a.
- There is one more function. x is equal to a power x.
- How is a power x written in terms of exponential function?
- can be written as e par log a par x.
- It is e par log.

- E per log a par x, that is e per log a into x.
- It is also a constant which is log a.
- Replace d by log a.
- V plus d by minus login if you have a bar minus.
- It is a non-homogeneous LDE.
- Why it is a non-homogeneous LDE?
- Because RHS is not equal to 0.
- What is the order of the differential equation?
- And this differential equation is a differential equation with constant coefficient.
- Solving this involves two major steps.
- The first step is to find the complementary function, and the second step is to find the particular integral.
- First, to find complementary function, I need to write the associated homogeneous differential equation of this differential equation.
- It is d square minus 2d plus 2 of y is equal to 0.
- And the auxiliary equation of the homogeneous equation is m square minus 2m plus 2 equal to 0.
- The roots are 1 plus r minus i because this is a quadratic equation, two roots, one root is 1 plus i and other root is 1 minus i.
- When the roots of the auxiliary equation are complex in nature.
- C1 cos what is beta here?
- The coefficient of i is 1.
- C1 cos 1x plus c2 sin x.
- Non-homogeneous differential equation is this.
- To find the particular integral.
- Particular integral is e power minus x.
- by d square minus 2t plus 2 plus e power x by d square minus 2t plus 2 because the RHS function is a power minus x plus e po x, which I split it into two terms.
- It's simple.

Example or application from the lecture

- Let us solve a problem.
- Y c the complementary function is also given the notation y c it is subscript c y c is equal to e power alpha x that is e to the power 1 into x into.
- The complementary function of the given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 20:00–24:59

replace d by e to the power a is one.

Key details

- Replace d by one.

- How do you simplify the first term?
- Replace d by minus log a.
- Minus 2 into minus log a plus 2.
- Plus e power x in this replace d by 1.
- equation is Y C plus 1.
- This is how we find out the particular integral of the non-homogeneous differential equation when capital X the RHS of the differential equation is an exponential function.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral of standard functions - Type 1 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 1 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 1 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9990; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 1 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

C Revision and retrieval practice